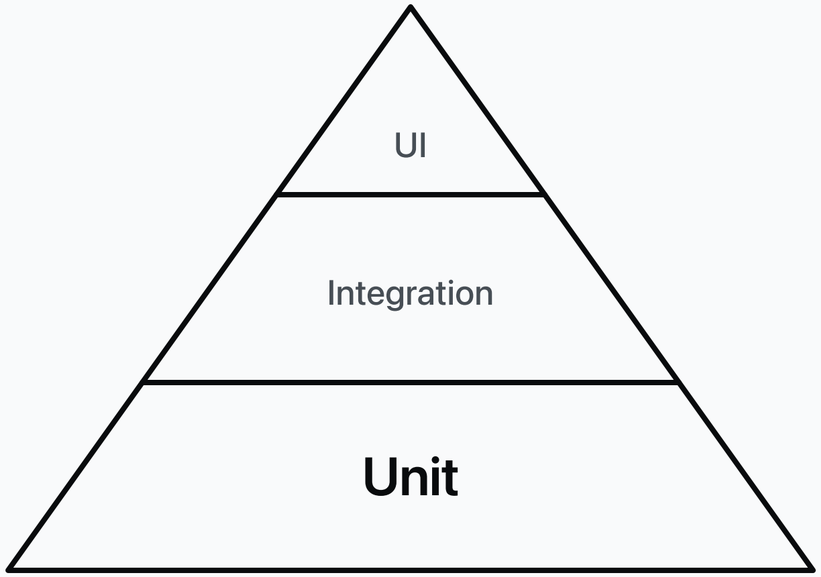
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Testing strategy

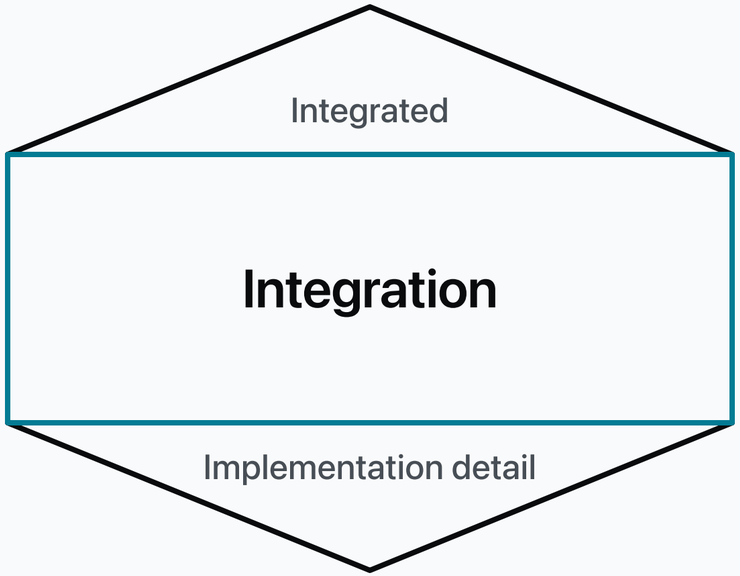
SmartAgents · Isabel · Ponto ops · October 2026

The shape of a test suite



The testing pyramid

Most tests at the bottom. Many unit tests, few above them.



The testing honeycomb

Most tests in the middle. Integration tests carry the weight.

Why the honeycomb

Behaviour, not functions	We are not asking whether a function or a line of code works. We are asking what the system does.
Preventing regression	A change that breaks behaviour that worked yesterday fails a test before it reaches production.
The real dependencies	Tests run against a real database and real libraries, where a mock would hide the fault.
Free to change the code	A test tied to behaviour stays green when the code underneath it is rewritten.

Before

99 FICs. Six of them were tested.

A small sample of six was chosen. For those six, about 18% of the code was covered.

After

100% covered	Every FIC is tested now, not a sample of six.
95 FICs remain	The integration tests proved that four were broken and never used.
Faulty production code	Code paths that contained faulty production code were discovered.
Unreachable code	Code that nothing can ever reach was found.

Everything is tested, every time.

It is one codebase, where overrides are made in config instead of a project per deployable. All the code is intertwined with if statements, and that cannot be untangled at short notice.

What only an integration test finds

The current CloudHSM depends on OpenSSL 1.0, not on 3.0.

Unit test	Integration test
CloudHSM is replaced by a mock	The real CloudHSM client, in a test container
The real library is never loaded	**The real library is loaded**
OpenSSL is never touched	**It asks for OpenSSL 1.0, not 3.0**
The dependency stays hidden	**The dependency shows before production**

The cost of covering all 95 FICs

01

150 seconds

Every FIC gets its own suite.
One whole run of one suite.

02

Almost 4 hours

All 95 suites, one after the other.
Too long to run every time.

03

20 minutes

Twelve suites at the same time on one laptop. Wall clock for all 95, and acceptable.

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Any questions?

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